

**DYNAMICAL SYSTEMS:
CLASS SEPTEMBER 24, 2026
9. TESTING FOR CLOSED ORBITS**

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Teams

Post screenshots of your work to the course Google Drive today: Link to drive folder. Include words, labels, and other short notes on your whiteboard that might make those solutions useful to you or your classmates. Name your images using the format:

- `teamXX-YY.EXT`

where XX is your two-digit team number (e.g. 06 if you are team 6), YY is the two digit number of image for today (e.g. 01 for the first image, 02 for the second, and so on.), and EXT is the extension for the image filetype you used.

Team Activity

PROBLEMS

(1) (Constructing a trapping region in polar)

Consider the system

$$\dot{r} = r(2 - \sin \theta - r), \quad \dot{\theta} = 1.$$

- (a) Argue that the curve $r = 2 - \sin \theta$ does not exactly correspond to a trajectory of the system.
- (b) For $r = 4$, identify the minimum and maximum possible values of $\dot{r}$ that could occur.
- (c) Construct a trapping region that satisfies the conditions of the Poincaré–Bendixson theorem.

Solution. (a) The curve

$$r = 2 - \sin \theta$$

is the radial nullcline, since on this curve

$$\dot{r} = r(2 - \sin \theta - r) = 0.$$

However, a trajectory lying on this curve would have to remain on it as θ changes. Differentiating $r = 2 - \sin \theta$ along such a curve gives

$$\dot{r} = -\cos \theta \dot{\theta} = -\cos \theta,$$

because $\dot{\theta} = 1$. Except at the isolated angles for which $\cos \theta = 0$, this is not zero. Thus the vector field is not tangent to the curve $r = 2 - \sin \theta$, so the curve is not itself a trajectory.

(b) At $r = 4$,

$$\dot{r} = 4(2 - \sin \theta - 4) = -8 - 4 \sin \theta.$$

Since $-1 \leq \sin \theta \leq 1$,

$$-12 \leq \dot{r} \leq -4.$$

Hence the minimum possible value is -12 and the maximum possible value is -4 . In particular, the vector field points strictly inward on the circle $r = 4$.

(c) Choose an annulus

$$0 < r_0 \leq r \leq 4$$

with, for example, $r_0 = \frac{1}{2}$. On the outer boundary $r = 4$, part (b) shows that $\dot{r} < 0$, so trajectories point into the annulus. On the inner boundary $r = \frac{1}{2}$,

$$\dot{r} = \frac{1}{2} \left(2 - \sin \theta - \frac{1}{2} \right) = \frac{1}{2} \left(\frac{3}{2} - \sin \theta \right) > 0,$$

since $\sin \theta \leq 1$. Thus trajectories also point into the annulus across its inner boundary. Therefore

$$\frac{1}{2} \leq r \leq 4$$

is a compact positively invariant trapping region. It excludes the equilibrium at $r = 0$, so it has the form needed for an application of the Poincaré–Bendixson theorem.

□

(2) Consider the system

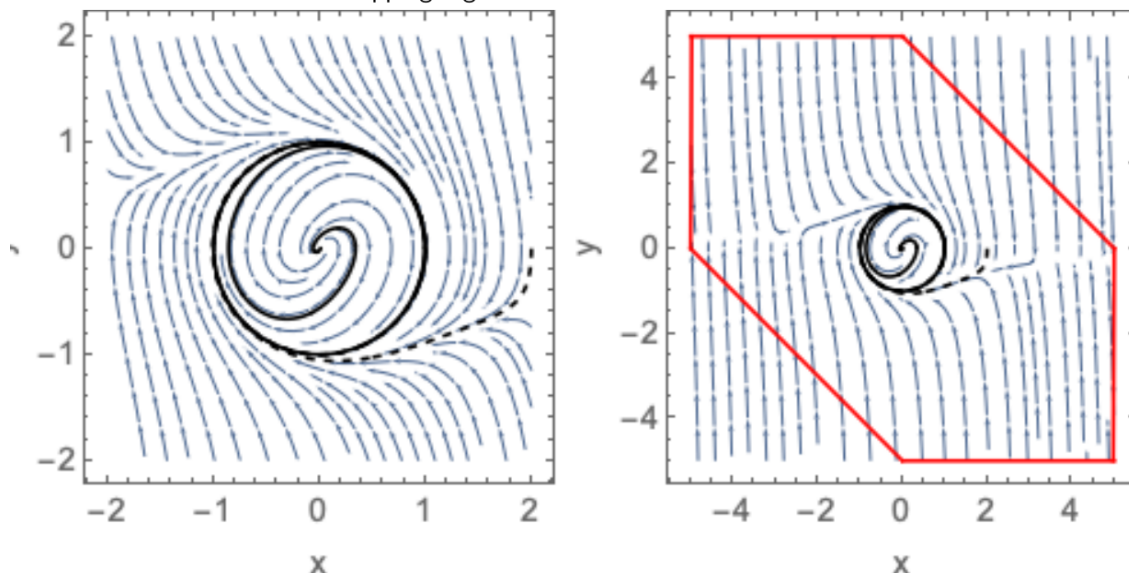
$$\dot{x} = y, \quad \dot{y} = -x - y(x^2 + y^2 - 1).$$

- (a) Transform the system to polar coordinates.
- (b) Show that there is an invariant set at

$$r = \sqrt{x^2 + y^2} = 1.$$

- (c) In polar coordinates the natural trapping regions are circles. If you try to create a circular outer trapping region, what issue arises?
- (d) Consider the box on the right, below. What mathematical work would you need to do to show that it is a trapping region?

(d) Consider the box on the right, below. What mathematical work would you need to do to show that it is a trapping region?



Solution. (a) Put $x = r \cos \theta$ and $y = r \sin \theta$. Using

$$\dot{r} = \frac{x\dot{x} + y\dot{y}}{r}, \quad \dot{\theta} = \frac{x\dot{y} - y\dot{x}}{r^2},$$

we obtain

$$\begin{aligned} \dot{r} &= \frac{xy + y[-x - y(r^2 - 1)]}{r} \\ &= -\frac{y^2}{r}(r^2 - 1) \\ &= r \sin^2 \theta (1 - r^2), \end{aligned}$$

and

$$\begin{aligned} \dot{\theta} &= \frac{x[-x - y(r^2 - 1)] - y^2}{r^2} \\ &= -1 - \frac{xy}{r^2}(r^2 - 1) \\ &= -1 - (r^2 - 1) \sin \theta \cos \theta. \end{aligned}$$

Thus

$$\dot{r} = r \sin^2 \theta (1 - r^2), \quad \dot{\theta} = -1 - (r^2 - 1) \sin \theta \cos \theta.$$

(b) On $r = 1$,

$$\dot{r} = r \sin^2 \theta (1 - r^2) = 0.$$

Hence there is no radial component of the vector field across the unit circle. Moreover, on $r = 1$,

$$\dot{\theta} = -1,$$

so the motion continues around the circle. Therefore $r = 1$ is invariant (indeed, it is a periodic orbit).

- (c) For every $r > 1$,

$$\dot{r} = r \sin^2 \theta (1 - r^2) \leq 0,$$

but at $\theta = 0$ and $\theta = \pi$ we have $\sin \theta = 0$, so $\dot{r} = 0$. Thus on any circular outer boundary the vector field is only weakly inward: at the two points on the x -axis it is tangent to the circle rather than pointing strictly inward. Consequently a simple strict inward-pointing test on a circular boundary does not work.

- (d) For the polygonal box shown, one must check the direction of the vector field on *each boundary segment*. If

$$F(x, y) = (y, -x - y(x^2 + y^2 - 1)),$$

and n denotes the outward normal to a side, the required calculation is

$$F(x, y) \cdot n \leq 0$$

at every point of that side (strictly < 0 wherever a strict trapping condition is desired).

From the picture the six sides may be represented as

$$\begin{aligned} y = 5, & & -5 \leq x \leq 0, & & y = x + 5, & & 0 \leq x \leq 5, \\ x = 5, & & -5 \leq y \leq 0, & & y = -5, & & 0 \leq x \leq 5, \\ y = -x - 5, & & -5 \leq x \leq 0, & & x = -5, & & 0 \leq y \leq 5. \end{aligned}$$

For example, on the top side $y = 5$ the outward normal is $(0, 1)$, so one checks

$$F \cdot n = \dot{y} = -x - 5(x^2 + 24) < 0.$$

On the right side $x = 5$ the outward normal is $(1, 0)$, and

$$F \cdot n = \dot{x} = y \leq 0.$$

On the upper-right diagonal $y = x + 5$, an outward normal can be chosen as $(-1, 1)$, so one checks

$$F \cdot (-1, 1) = -\dot{x} + \dot{y} \leq 0,$$

throughout that segment. The analogous dot-product inequalities are checked on the remaining three sides. Establishing all six inequalities proves that no trajectory can cross the polygon outward, and hence that the box is a trapping region.

□

Liapunov functions:

A Liapunov function for a system is a continuously differentiable function $V(x, y)$ with the following properties:

1. $V(x, y) > 0$ for all (x, y) not equilibria,
2. If (x, y) is an equilibrium point, then $V(x, y) = 0$,
3. $\dot{V} < 0$ along all trajectories to an equilibrium.

The existence of a Lyapunov function on a region U allows us to rule out closed trajectories in that region.

Theorem: If L is a Liapunov function for an equilibrium $\mathbf{x}$ then the equilibrium is asymptotically stable: every point in U will approach $\mathbf{x}^*$ as $t \rightarrow \infty$.

Bendixson's criterion is a method for ruling out closed trajectories on some regions of a plane.

On regions where $f_x + g_y$ has a single sign, a closed trajectory is not possible.

Justification: The flux of a vector field across a closed trajectory is zero. The vector field is tangent to the closed trajectory at each point along the trajectory, so $\mathbf{F} \cdot \hat{\mathbf{n}} = 0$ at every point along such a curve.

The divergence form of Green's theorem states that

$$\oint_{\partial R} \mathbf{F} \cdot \hat{\mathbf{n}} ds = \iint_R \nabla \cdot \mathbf{F} dA,$$

where R is a region in the plane and $C = \partial R$ is its boundary curve. This theorem states that the integral of the divergence over a region is equal to the flux out the boundary of the region. If

$$\iint_R \nabla \cdot \mathbf{F} dA \neq 0,$$

then

$$\oint_{\partial R} \mathbf{F} \cdot \hat{\mathbf{n}} ds \neq 0$$

and the boundary of R is not a closed trajectory.

(3) **(Ruling out closed orbits)** Let

$$\dot{x} = y, \quad \dot{y} = x - x^3 - \mu y, \quad \mu \geq 0.$$

This system is known as the unforced Duffing oscillator. (Duffing published work on this kind of equation in 1918, and it became a popular model in the 1970s). The μy term is a damping term.

(a) This system can also be written as

$$\ddot{x} = x - x^3 - \mu \dot{x}.$$

How does this differ from the similar form that we have seen for a conservative system?

(b) Use Bendixson's criterion to show that the system has no closed orbits for $\mu > 0$.

Solution. (a) Since $y = \dot{x}$, differentiating gives $\dot{y} = \ddot{x}$. Hence

$$\ddot{x} = x - x^3 - \mu \dot{x},$$

or equivalently

$$\ddot{x} + \mu \dot{x} - x + x^3 = 0.$$

A conservative equation has the form

$$\ddot{x} = -V'(x)$$

with no velocity-dependent damping term. Here the term $-\mu \dot{x}$ dissipates energy when $\mu > 0$, so the system is not conservative unless $\mu = 0$.

(b) Let

$$f(x, y) = y, \quad g(x, y) = x - x^3 - \mu y.$$

Its divergence is

$$f_x + g_y = 0 - \mu = -\mu.$$

If $\mu > 0$, then

$$f_x + g_y = -\mu < 0$$

everywhere in the plane. Since the divergence has one strict sign on the simply connected plane, Bendixson's criterion rules out every nonconstant closed orbit. Thus the damped Duffing oscillator has no periodic orbit for $\mu > 0$. □

- (4) Find a restriction on a and b such that

$$V(x, y) = ax^2 + by^2$$

is a Liapunov function for

$$\dot{x} = y - x^3, \quad \dot{y} = -x - y^3.$$

Solution. We have

$$V(x, y) = ax^2 + by^2,$$

so V is positive definite provided

$$a > 0, \quad b > 0.$$

Along trajectories,

$$\begin{aligned} \dot{V} &= 2ax\dot{x} + 2by\dot{y} \\ &= 2ax(y - x^3) + 2by(-x - y^3) \\ &= 2(a - b)xy - 2ax^4 - 2by^4. \end{aligned}$$

The mixed quadratic term $2(a - b)xy$ has no fixed sign. To guarantee that $\dot{V} < 0$ for every $(x, y) \neq (0, 0)$, it must vanish, so we require

$$a = b.$$

With $a = b > 0$,

$$V = a(x^2 + y^2) > 0 \quad ((x, y) \neq (0, 0))$$

and

$$\dot{V} = -2a(x^4 + y^4) < 0 \quad ((x, y) \neq (0, 0)).$$

Therefore the required restriction is

$$a = b > 0.$$

□